

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Trigonometric Identities

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Simplify $(1 - \cos^2 A)/\sin A$.

Answer:

Use the standard identity and simplify the expression to obtain $\sin A$.

Question 2

Simplify $(1 + \tan^2 A)\cos^2 A$.

Answer:

Use the standard identity and simplify the expression to obtain 1.

Question 3

Simplify $(\sec^2 A - 1)/\tan A$.

Answer:

Use the standard identity and simplify the expression to obtain $\tan A$.

Question 4

Simplify $\sin^2 A/(1 - \cos^2 A)$.

Answer:

Use the standard identity and simplify the expression to obtain 1.

Question 5

Simplify $(1 - \cot^2 A)/(1 - \operatorname{cosec}^2 A)$.

Answer:

Use the standard identity and simplify the expression to obtain $1 - \tan^2 A$.

Question 6

Simplify $(\sin A)/(1 + \cos A) + (1 + \cos A)/\sin A$.

Answer:

Use the standard identity and simplify the expression to obtain $2 \operatorname{cosec} A$.

Question 7

Simplify $(\sec A - \tan A)(\sec A + \tan A)$.

Answer:

Use the standard identity and simplify the expression to obtain 1.

Question 8

Simplify $(1 + \sin A)(1 - \sin A)$.

Answer:

Use the standard identity and simplify the expression to obtain $\cos^2 A$.

Question 9

Simplify $\tan A/(\sec A + 1)$.

Answer:

Use the standard identity and simplify the expression to obtain $(\sec A - 1)/\tan A$.

Trigonometric Identities - Continued

Question 10

Simplify $(\operatorname{cosec}^2 A - 1)/\cot A$.

Answer:

Use the standard identity and simplify the expression to obtain $\cot A$.

Question 11

If $\sec A = 13/12$ for an acute angle, find $\tan A$.

Answer:

$\tan^2 A = \sec^2 A - 1 = (13/12)^2 - 1$. Hence $\tan A = 5/12$.

Question 12

If $\sec A = 5/4$ for an acute angle, find $\tan A$.

Answer:

$\tan^2 A = \sec^2 A - 1 = (5/4)^2 - 1$. Hence $\tan A = 3/4$.

Question 13

If $\sec A = 17/8$ for an acute angle, find $\tan A$.

Answer:

$\tan^2 A = \sec^2 A - 1 = (17/8)^2 - 1$. Hence $\tan A = 15/8$.

Question 14

If $\sec A = 25/7$ for an acute angle, find $\tan A$.

Answer:

$\tan^2 A = \sec^2 A - 1 = (25/7)^2 - 1$. Hence $\tan A = 24/7$.

Question 15

If $\sec A = 10/6$ for an acute angle, find $\tan A$.

Answer:

$\tan^2 A = \sec^2 A - 1 = (10/6)^2 - 1$. Hence $\tan A = 8/6$.

Question 16

If $\tan A = 3/4$, find $\sec A$.

Answer:

Take opposite=3, adjacent=4; hypotenuse=5. Thus $\sec A = 5/4$.

Question 17

If $\tan A = 5/12$, find $\sec A$.

Answer:

Take opposite=5, adjacent=12; hypotenuse=13. Thus $\sec A = 13/12$.

Question 18

If $\tan A = 8/15$, find $\sec A$.

Answer:

Take opposite=8, adjacent=15; hypotenuse=17. Thus $\sec A = 17/15$.

Question 19

If $\tan A = 7/24$, find $\sec A$.

Answer:

Take opposite=7, adjacent=24; hypotenuse=25. Thus $\sec A = 25/24$.

Trigonometric Identities - Continued

Question 20

If $\tan A = 9/40$, find $\sec A$.

Answer:

Take opposite=9, adjacent=40; hypotenuse=41. Thus $\sec A = 41/40$.

Question 21

State the main formula, rule or theorem used in Trigonometric Identities.

Answer:

$\sin^2 A + \cos^2 A = 1$, $1 + \tan^2 A = \sec^2 A$ and $1 + \cot^2 A = \operatorname{cosec}^2 A$.

Question 22

What do the important symbols or quantities in Trigonometric Identities represent?

Answer:

An identity is true for every angle where both sides are defined.

Question 23

Write the correct method for solving a Trigonometric Identities numerical.

Answer:

Start with one side, replace a part using a standard identity, and simplify without changing both sides at once.

Question 24

Which unit, sign convention or condition must be checked in a Trigonometric Identities answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Trigonometric Identities be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Trigonometric Identities questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Trigonometric Identities numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Trigonometric Identities?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Trigonometric Identities - Continued

Question 29

How can a student decide which formula to use in Trigonometric Identities?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Trigonometric Identities?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

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